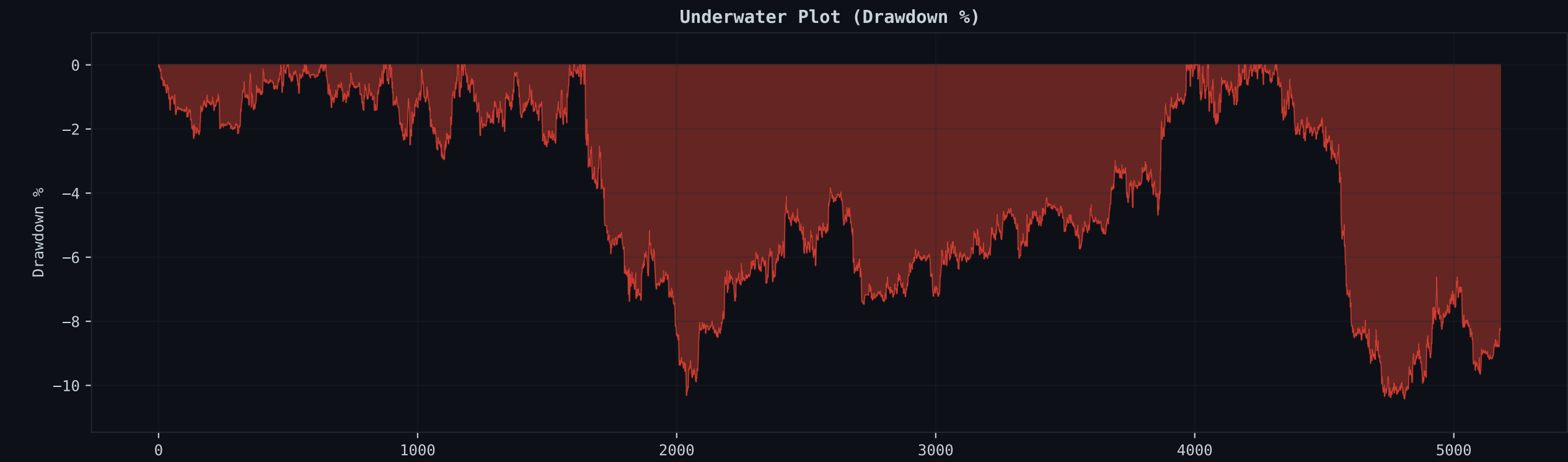
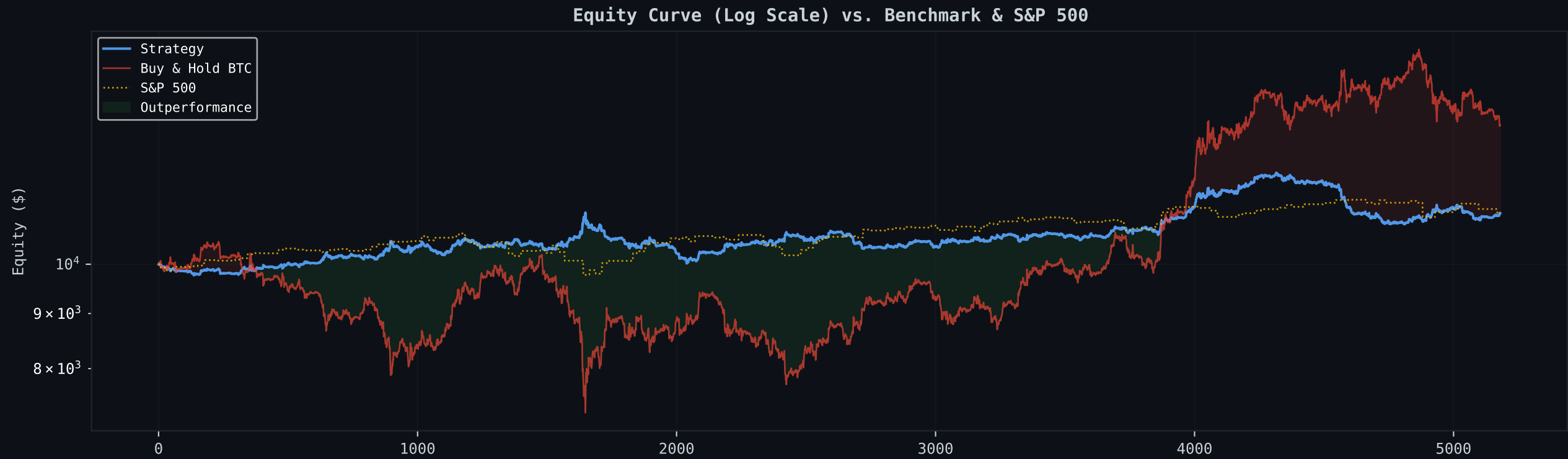
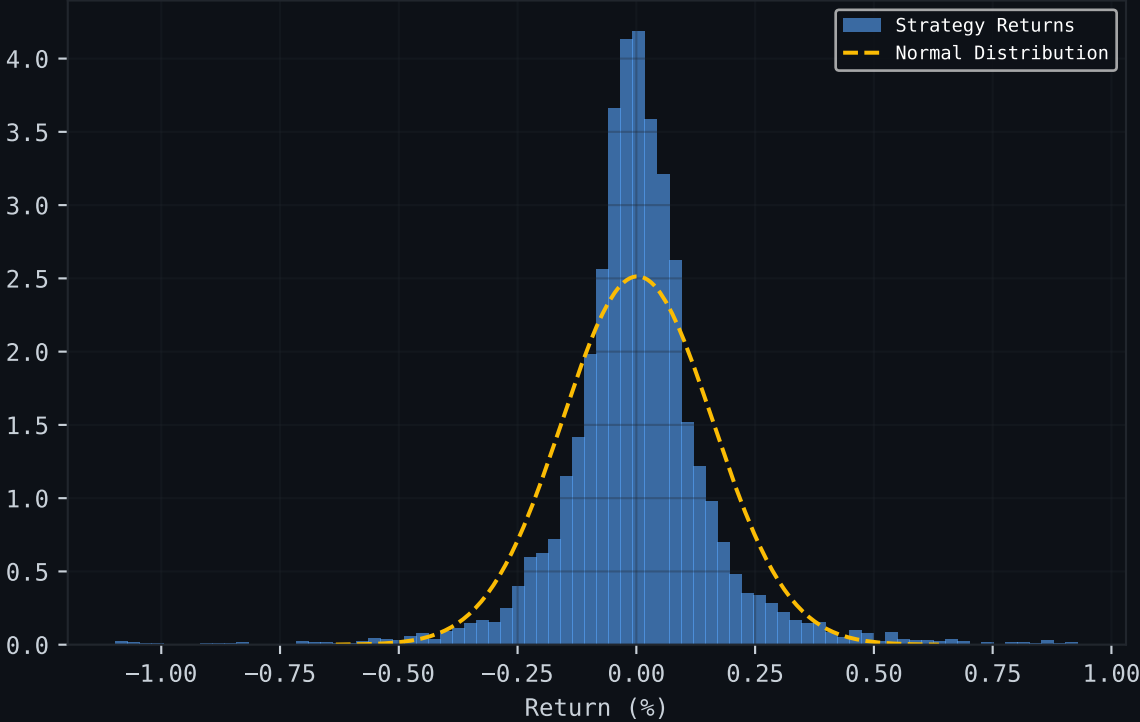


INSTITUTIONAL QUANTITATIVE TEARSHEET
CNN+LSTM PPO × SPA Day Trading Strategy

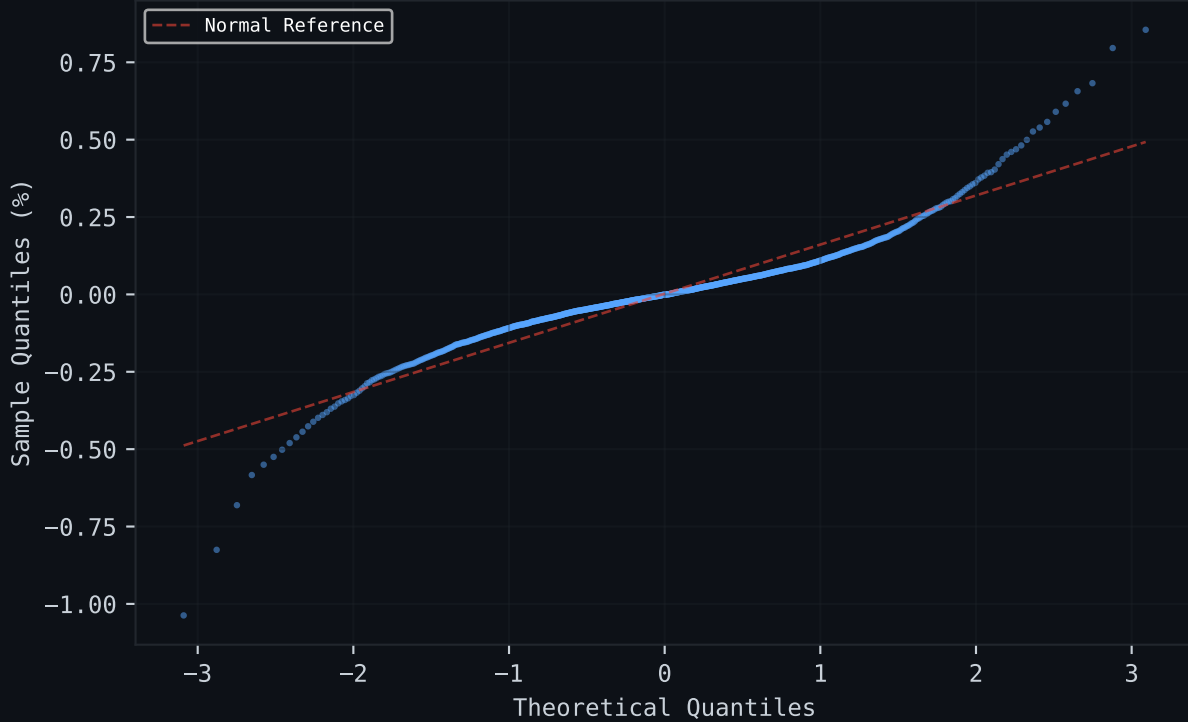
Sharpe: 1.36 | Sortino: 1.86 | Calmar: 1.95 | Max DD: -10.4% | Alpha p-value: 0.1482 | Beta: -0.041



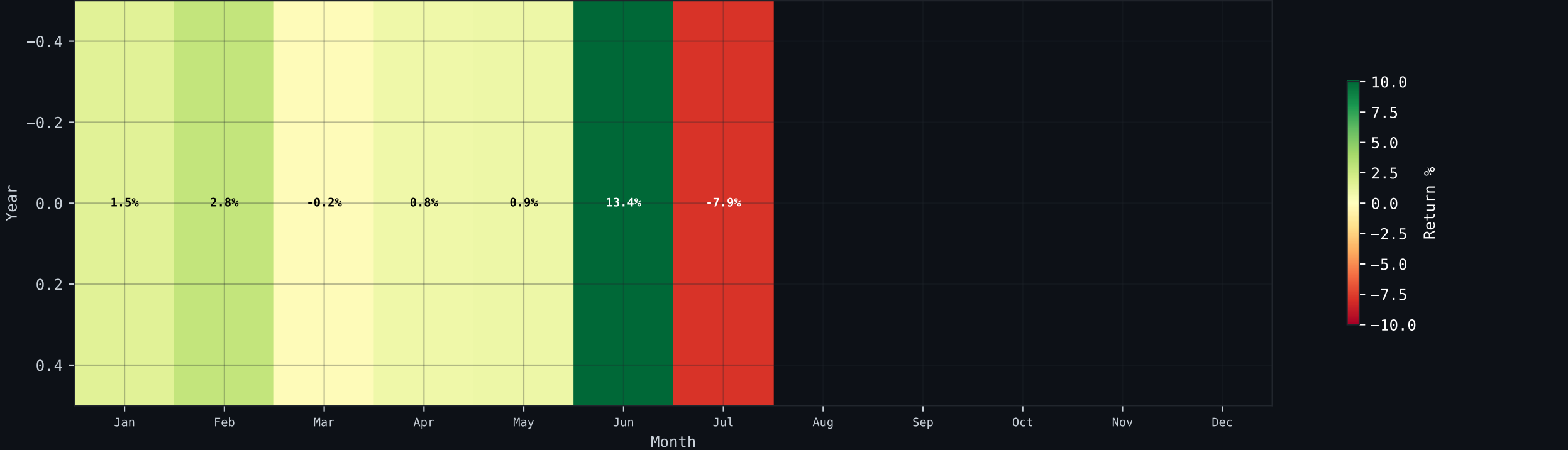
Return Distribution vs. Normal



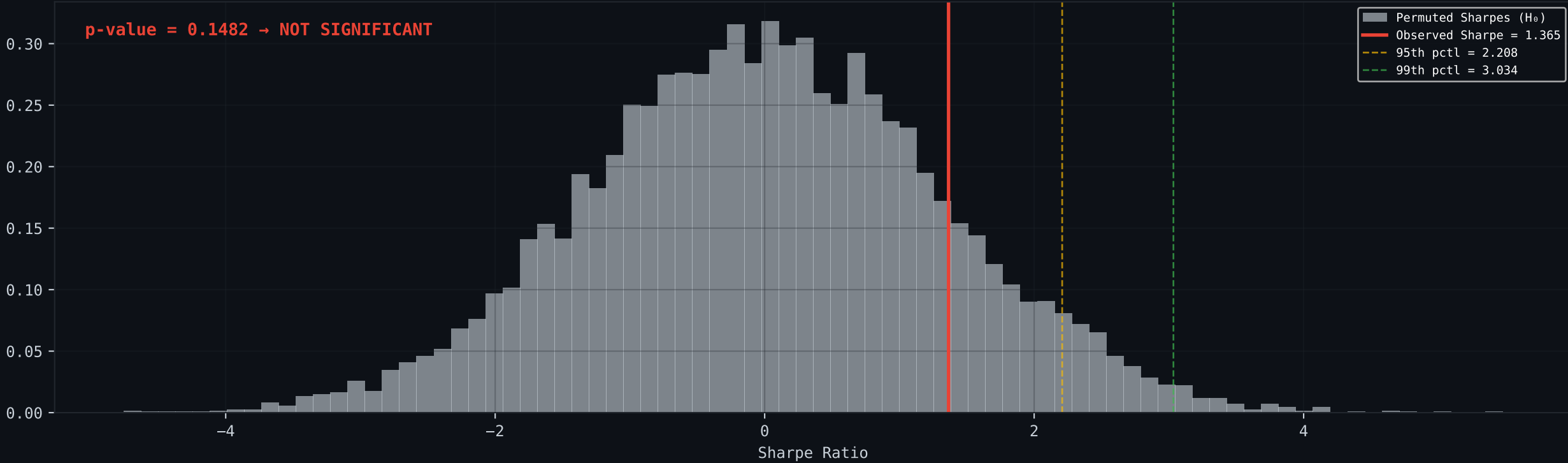
Q-Q Plot (Fat Tail Detection)



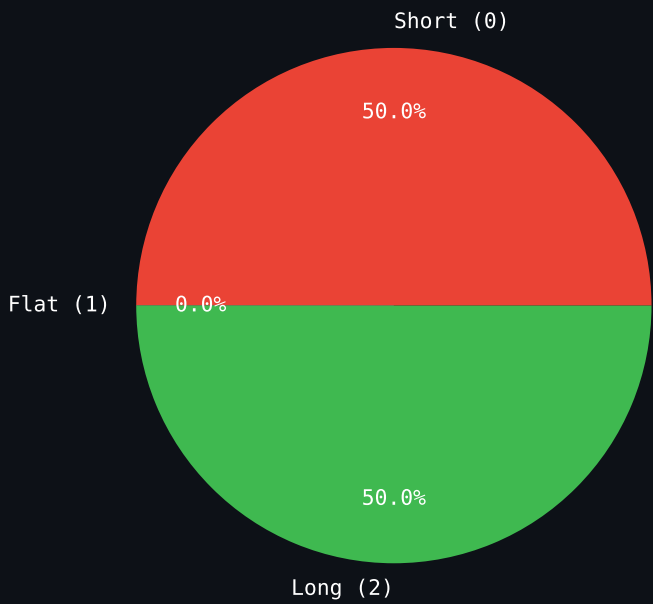
Monthly Returns Heatmap (%)



Monte Carlo Permutation Test (10,000 simulations)



Action Distribution



Trade Returns vs. Duration



QUANTITATIVE RISK METRICS – FULL DETAIL

Metric	Value
PERFORMANCE METRICS	
Total Return	11.53%
Annualized Return	20.27%
Annualized Volatility	14.86%
Sharpe Ratio	1.365
Sortino Ratio	1.857
Calmar Ratio	1.945
Omega Ratio (θ=0)	1.044
Profit Factor	1.044
Win Rate	49.4%
BENCHMARK COMPARISON	
BTC Buy & Hold Sharpe	1.294
BTC Buy & Hold Sortino	1.706
BTC Buy & Hold Max DD	-30.61%
S&P 500 Sharpe	1.468
S&P 500 Sortino	0.305
S&P 500 Max DD	-8.49%
TAIL RISK METRICS	
Max Drawdown	-10.42%
VaR (95%)	-0.2280%
VaR (99%)	-0.4436%
CVaR / Expected Shortfall (95%)	-0.3670%
CVaR / Expected Shortfall (99%)	-0.6411%
Skewness	0.007
Excess Kurtosis	7.245
ALPHA DECOMPOSITION	
Alpha (per period)	0.000025
Alpha t-statistic	1.159
Alpha p-value	0.2466
Beta (benchmark exposure)	-0.0413
R² (benchmark correlation)	0.0197
STATISTICAL VALIDATION	
Permutation Test p-value	0.1482
Significance (p < 0.05)	NO ☐
CAPACITY ESTIMATION	
Est. Capacity (USD)	\$18,622,735,424,652
Current Market Impact	0.33 bps
Participation Rate	0.000011%